1. Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

To divide the network **192.168.10.0/24 into 4 equal subnets**, we need to borrow 2 bits from the host portion.

Step 1: Calculate the New Subnet Mask

- **Original network:** 192.168.10.0/24
- **Number of subnets required:** 4
- **Borrowed bits:** 2 (because $2^2 = 4$)
- **New subnet prefix:** /26
- **New subnet mask:** 255.255.255.192

Each subnet contains 64 IP addresses, including the network and broadcast addresses, with **62 usable host addresses**.

Step 2: Identify the Four Subnets

Subnet	Network Address	Subnet Mask	Broadcast Address	Valid Host Range
1	192.168.10.0/26	255.255.255.192	192.168.10.63	192.168.10.1 – 192.168.10.62
2	192.168.10.64/26	255.255.255.192	192.168.10.127	192.168.10.65 – 192.168.10.126
3	192.168.10.128/26	255.255.255.192	192.168.10.191	192.168.10.129 – 192.168.10.190
4	192.168.10.192/26	255.255.255.192	192.168.10.255	192.168.10.193 – 192.168.10.254

Step 3: Understand the Subnet Increments

The subnet increment is calculated as:

256 − 192 = 64

Therefore, the network addresses increase by 64:

- Subnet 1: 192.168.10.0
- Subnet 2: 192.168.10.64
- Subnet 3: 192.168.10.128
- Subnet 4: 192.168.10.192

Final Answer

The network **192.168.10.0/24** is divided into 4 equal subnets using the **/26 prefix (255.255.255.192)**. Each subnet contains 64 total IP addresses, including 62 usable host addresses, one network address, and one broadcast address.

2. For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

Hint : Make sure not to use the network or broadcast address for any device.

Let's use Subnet 1: 192.168.10.0/27

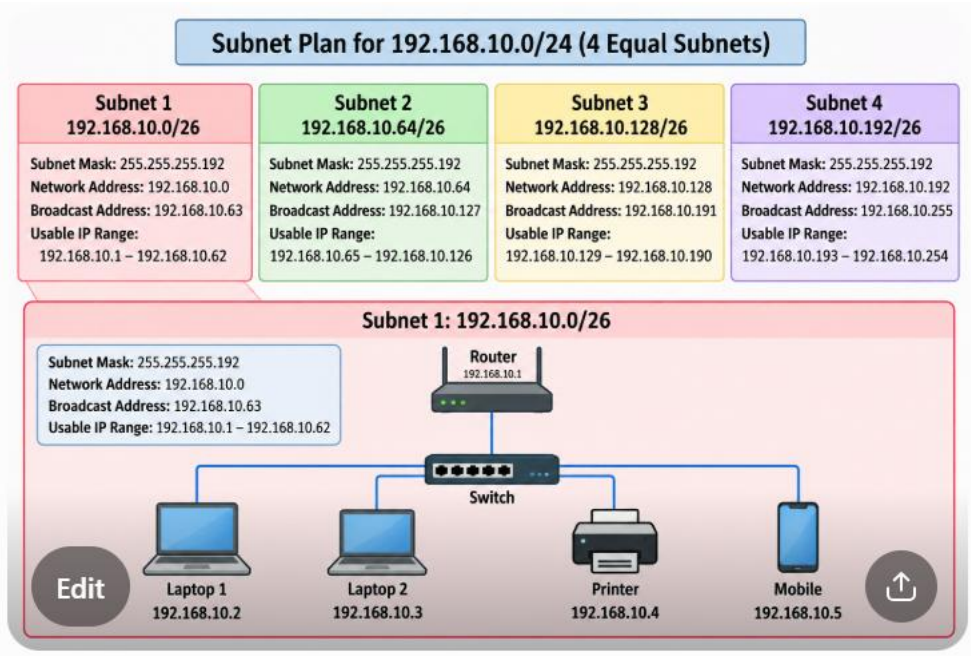
- **Network Address:** 192.168.10.0
- **Subnet Mask:** 255.255.255.224
- **Broadcast Address:** 192.168.10.31
- **Valid IP Range:** 192.168.10.1 – 192.168.10.30

IP Addresses Assigned to 5 Devices

Device	Assigned IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile	192.168.10.5

Conclusion: All five devices have valid IP addresses within the subnet. The network address (192.168.10.0) and broadcast address (192.168.10.31) are not assigned to any device.

3. Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.



4. Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Step 1: Identify the Network

My original network address is 192.168.10.0/24. I divided it into 4 equal subnets.

Step 2: Calculate the Subnet Mask

To create 4 subnets, I borrowed 2 bits because $2^2 = 4$.

- **Original prefix:** /24
- **Borrowed bits:** 2
- **New prefix:** /26 for 4 equal subnets.

Subnet mask: 255.255.255.192

Step 3: Calculate the Number of Hosts

I used the formula:

Number of usable hosts = $2^n - 2$

Here, *n* is the number of host bits.

- Host bits = $32 - 26 = 6$
- Total IP addresses = $2^6 = 64$
- Usable host addresses = $64 - 2 =$ **62**

Each subnet has 62 usable host addresses.

Step 4: Calculate the Network and Broadcast Addresses

The subnet increment is:

$256 - 192 =$ **64**

Therefore, the four subnets are:

Subnet	Network Address	Broadcast Address
1	192.168.10.0	192.168.10.63
2	192.168.10.64	192.168.10.127
3	192.168.10.128	192.168.10.191
4	192.168.10.192	192.168.10.255

Step 5: Assign IP Addresses to Devices

I selected Subnet 1, which has the network address 192.168.10.0/26.

- Network address: 192.168.10.0
- Broadcast address: 192.168.10.63
- Valid IP range: 192.168.10.1 to 192.168.10.62

I assigned the following IP addresses to five devices:

Device	Assigned IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile	192.168.10.5

Step 6: Check the IP Addresses

I checked that all five devices have IP addresses within the valid range. I did not assign the network address or broadcast address to any device.

Conclusion

I divided the original network into four equal subnets using the subnet mask **255.255.255.192 (/26)**. Each subnet has 64 total IP addresses and 62 usable host addresses. I assigned valid IP addresses to the router, two laptops, a printer, and a mobile phone in Subnet 1.